SOLUTIONS FOR WORKSHEET 2

Question 1. Using the definition of a convergent sequence to prove the following.

(a) $\lim_{n\to\infty}\left(5-\frac{3}{n}\right)=5$ (b) $\lim_{n\to\infty}\frac{n+4}{n+6}=1$ (c) $\lim_{n\to\infty}\frac{3\cos n}{n+1}=0$ (d) $\lim_{n\to\infty}\frac{7\sin n}{n}=0$

(e) $\lim_{n\to\infty} c = c$, c is a real constant.

Proof. Let $k>0$. By Archimedean property, there exists a natural number N such that $\frac{\varepsilon}{k}>\frac{1}{N}$ implies that $\varepsilon>\frac{k}{N}$ for all $\varepsilon\in\mathbb{R}^+$.

(a) $n>N \Rightarrow \left|5-\frac{3}{n}-5\right| = \frac{3}{n} < \frac{3}{N} < \varepsilon$ (choose $k=3$).

(b) $n>N \Rightarrow \left|\frac{n+4}{n+6}-1\right| = \frac{2}{n+6} < \frac{2}{n} < \frac{2}{N} < \varepsilon$ (choose $k=2$).

(c) $n>N \Rightarrow \left|\frac{3\cos n}{n+1}-0\right| \le \frac{3}{n+1} < \frac{3}{n} < \frac{3}{N} < \varepsilon$ (choose $k=3$),

Note that $\left|\cos\theta\right| \le 1$ for all $\theta\in\mathbb{R}$.

(d) $n>N \Rightarrow \left|\frac{7\sin n}{n}-0\right| \le \frac{7}{n} < \frac{7}{N} < \varepsilon$ (choose $k=7$),

Note that $\left|\sin\theta\right| \le 1$ for all $\theta\in\mathbb{R}$.

(e) $\left|c-c\right| = 0 < \frac{1}{n} < \varepsilon$ for all $n\in\mathbb{N}$ (choose $k=1$, $n=N$).

Question 2. Give an example or prove the following is impossible.

(a) A sequence with an infinite number of zeros that does not converge to zero.

(b) A sequence with an infinite number of zeros that converges to a non-zero limit.

(c) A divergent sequence such that for all $n\in\mathbb{N}$, there are n consecutive zeros somewhere in the sequence.

Answer. (a) $\left(1+(-1)^n\right)$.

(b) Let (a_n) be a sequence as described in (b), and it converges to $a\neq 0$. Then for all $\varepsilon>0$, there exists a natural number N such that $|a_n-a|<\varepsilon$ for all $n>N$. Choose $\varepsilon=\frac{|a|}{2}>0$ because $a\neq 0$. Now, there exists an $n>M$ such that $a_n=0$. So, $|a_n-a|=|a|<\frac{|a|}{2}$ for all $n>\max\{N, M\}$, which is a contradiction.

(c) $(1,0,1,0,0,1,0,0,0,1,0,0,0,0,1,\ldots)$.

Question 3. Prove $\lim\left(7-\frac{1}{\sqrt{n}}\right)=7$ by using the definition of a convergent sequence.

Proof. By Archimedean property, for all positive real number ε, there exists a natural number M such that $\varepsilon > 1/M$. Let $N = M^2$. Then $n > N = M^2 \Rightarrow \sqrt{n} > M$ *, which implies that $\frac{1}{\sqrt{n}} < \frac{1}{M}$. Hence, $\left|7-\frac{1}{\sqrt{n}}-7\right| = \frac{1}{\sqrt{n}} < \frac{1}{M} < \varepsilon$ for all $n > N$.

$*\, n - M^2 > 0 \Rightarrow \left(\sqrt{n}-M\right)\left(\sqrt{n}+M\right) > 0 \Rightarrow \sqrt{n}-M > 0$ as $n, M \in \mathbb{N} \Rightarrow \sqrt{n}+M > 0$.

Question 4. Prove or disprove: If (a_n) converges, then $\left\{a_n : n \in \mathbb{N}\right\}$ has a maximum.

Proof. Sequence $\left(1-\frac{1}{n}\right)$ converges but $\max\left\{1-\frac{1}{n} : n \in \mathbb{N}\right\}$ does not exist.

Question 5. Suppose that $\lim_{n\to\infty} a_n = a$. Prove that

(a) $\lim_{n\to\infty}\left(5a_n+3\right)=5a+3$ (b) $\lim_{x\to\infty}\left|a_n\right|=\left|a\right|$ (Hint: use $\left|\left|a\right|-\left|b\right|\right| \le \left|a-b\right|\ \forall a,b \in \mathbb{R}$.)

(c) $\lim_{n\to\infty}\left(2a_{n+1}+3a_n\right)=5a$ (d) $\lim_{n\to\infty}\left(a_{n+2}-2a_{n+1}+a_n\right)=0$

Proof. Note: Let k be a fixed positive real number. For all $\frac{\varepsilon}{k} > 0$, there exists $N \in \mathbb{N}$ such that $\left|a_n - a\right| < \frac{\varepsilon}{k}$ whenever $n > N$.

(a) $n > N \Rightarrow \left|5a_n + 3 - (5a+3)\right| = 5\left|a_n - a\right| < \varepsilon$ (choose $k = 5$).

(b) $n > N \Rightarrow \left|\left|a_n\right| - \left|a\right|\right| \le \left|a_n - a\right| < \varepsilon$ (choose $k = 1$).

(c) Note that $\lim_{n\to\infty} a_n = a \Rightarrow \lim_{n\to\infty} a_{n+1} = a$. To prove this, we let $u = n+1$, then

$$\lim_{n\to\infty} a_{n+1} = \lim_{u-1\to\infty} a_u = \lim_{u\to\infty} a_u = a.$$

Thus, for any given $\varepsilon > 0$, there exist $N_1, N_2 \in \mathbb{N}$ such that

$$\left|a_{n+1}-a\right| < \tfrac{\varepsilon}{4} \text{ for all } n > N_1 \text{ and } \left|a_n - a\right| < \tfrac{\varepsilon}{6} \text{ for all } n > N_2.$$

$$\therefore \left|2a_{n+1}+3a_n-5a\right| = \left|2a_{n+1}-2a+3a_n-3a\right|$$
$$\le 2\left|a_{n+1}-a\right| + 3\left|a_n-a\right| < 2\left(\tfrac{\varepsilon}{4}\right) + 3\left(\tfrac{\varepsilon}{6}\right) = \varepsilon.$$

Then $n > \max\{N_1, N_2\} \Rightarrow \left|2a_{n+1}+3a_n-5a\right| < \varepsilon$.

(d) Let $\varepsilon > 0$, there exist a positive integer N_1, N_2 and N_3 such that

$$n > N_1 \Rightarrow \left|a_{n+2}-a\right| < \tfrac{\varepsilon}{3},\ n > N_2 \Rightarrow \left|a_{n+1}-a\right| < \tfrac{\varepsilon}{6} \text{ and } n > N_3 \Rightarrow \left|a_n - a\right| < \tfrac{\varepsilon}{3}.$$

$$\left|a_{n+2}-2a_{n+1}+a_n\right| = \left|(a_{n+2}-a)+(2a-2a_{n+1})+(a_n-a)\right|$$
$$\le \left|a_{n+2}-a\right| + 2\left|a-a_{n+1}\right| + \left|a_n-a\right| < \tfrac{\varepsilon}{3} + 2\left(\tfrac{\varepsilon}{6}\right) + \tfrac{\varepsilon}{3} = \varepsilon$$

Then $n > \max\{N_1, N_2, N_3\} \Rightarrow \left|a_{n+2}-2a_{n+1}+a_n\right| < \varepsilon$.

Question 6. Let $K>0$. Let (a_n) and (b_n) be sequences with $\lim_{n\to\infty} b_n = 0$ and $|a_n| \le K|b_n|$ for all $n \in \mathbb{N}$. Prove from the definition of convergence of sequence that $\lim_{n\to\infty} a_n = 0$.

Proof. Since $\lim_{n\to\infty} b_n = 0$, it follows that for all $\frac{\varepsilon}{K} > 0$, $K>0$, that there exists $N \in \mathbb{N}$ such that $n > N \Rightarrow |b_n - 0| < \frac{\varepsilon}{K}$. Then for all $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that $n > N \Rightarrow |a_n - 0| = |a_n| \le K|b_n| < \varepsilon$.

Question 7. Let (a_n) be a sequence with $\lim_{n\to\infty} a_n = a$. Prove that there exists $N \in \mathbb{N}$ such that $|a_n| < 1 + |a| \ \ \forall n > N$.

Proof. Given that for all $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that $|a_n - a| < \varepsilon$ for all $n > N$. Choose $\varepsilon = 1$, then $|a_n| = |a_n - a + a| \le |a_n - a| + |a| < 1 + |a|$.

Question 8. Let (a_n) be a sequence with $\lim_{n\to\infty} a_n = a > 0$. Prove that

(a) there exists $N \in \mathbb{N}$ such that $a_n > \frac{1}{2}a$ for all $n > N$

(b) $\lim_{n\to\infty} \frac{1}{a_n} = \frac{1}{a}$

Proof. Given that for all $\tilde{\varepsilon} > 0$, there exists $N \in \mathbb{N}$ such that $|a_n - a| < \tilde{\varepsilon}$ for all $n > N$, where $a > 0$.

(a) Choose $\tilde{\varepsilon} = \frac{a}{2}$, then $|a_n - a| < \tilde{\varepsilon} = \frac{a}{2} \Rightarrow \frac{a}{2} < a_n < \frac{3a}{2}$.

(b) Let $\tilde{\varepsilon} = \frac{a^2\varepsilon}{2}$. Then from (a), $|a_n| \ge a_n > \frac{a}{2} \Rightarrow \frac{1}{|a_n|} < \frac{2}{a}$, so $\left|\frac{1}{a_n} - \frac{1}{a}\right| = \left|\frac{a-a_n}{aa_n}\right| < \frac{2\tilde{\varepsilon}}{a^2} = \varepsilon$.

Question 9. Prove that if (a_n) and (b_n) are sequences that both diverge to ∞, then $(a_n + b_n)$ also diverges to ∞.

Proof. For all $M \in \mathbb{R}^+$, there exists $N_1, N_2 \in \mathbb{N}$ such that

$$n > N_1 \Rightarrow a_n > M \ \text{ and } \ n > N_2 \Rightarrow a_n > M.$$

Let $N = \max\{N_1, N_2\}$. Then $n > N \Rightarrow a_n + b_n > 2M > M$.

Question 10. Using the definition of a convergent sequence, prove that

(a) $\lim_{n\to\infty} \dfrac{(-1)^n}{3n^2} = 0$

(b) $\lim_{n\to\infty} \dfrac{2n-3}{n^2 - 2n - 5} = 0$

(c) $\lim_{n\to\infty} \dfrac{n^2 + 5n - 2}{(n+3)^2 + (-1)^n + 2} = 1$

(d) $\lim_{n\to\infty} \dfrac{2 + ne^{-n}}{3n^2 + \sin n + 1} = 0$

Proof. (a) Archimedean property: For all $3\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that $3\varepsilon > \frac{1}{N}$. Then for all positive real number ε, there exists a $N \in \mathbb{N}$ such that

$$n > N \Rightarrow \left|\frac{(-1)^n}{3n^2} - 0\right| \le \frac{1}{3n^2} < \frac{1}{3N} < \varepsilon \text{. Note: } n^2 \ge n > N \Rightarrow \frac{1}{n^2} < \frac{1}{N}$$

(b) Archimedean property: For all $\frac{\varepsilon}{4} > 0$, there exists $N \in \mathbb{N}$ such that $\varepsilon > \frac{4}{N}$. Then for all positive real number ε, there exists a $N \in \mathbb{N}$ such that

$$n > N > 5 \Rightarrow \left|\frac{2n-3}{n^2-2n-5} - 0\right| < \frac{2n}{\frac{1}{2}n^2} = \frac{4}{n} < \frac{4}{N} < \varepsilon,$$

since $\frac{1}{2}n^2 < n^2 - 2n - 5 \ \forall n > 5$. Note: $n^2 - 2n - 5 > \frac{1}{2}n^2 \Rightarrow n > 2 + \sqrt{14}$.

(c) Archimedean property: For all $\frac{\varepsilon}{15} > 0$ there exists $N \in \mathbb{N}$ such that $\varepsilon > \frac{15}{N}$. Then

$$\left|a_n - 1\right| = \left|\frac{n + 13 + (-1)^n}{(n+3)^2 + (-1)^n + 2}\right| < \left|\frac{n + 13 + (-1)^n}{n^2}\right| < \left|\frac{n + 13n + n}{n^2}\right| = \frac{15}{n} < \frac{15}{N} < \varepsilon$$

whenever $n > \max\{N, 2\}$.

(d) Archimedean property: For all $\frac{3\varepsilon}{2} > 0$, there exists $N \in \mathbb{N}$ such that $\varepsilon > \frac{2}{3N}$.

$$n > \max\{N,2\} \Rightarrow \left|\frac{2 + ne^{-n}}{3n^2 + \sin n + 1}\right| < \frac{2+n}{3n^2} \le \frac{2n}{3n^2} = \frac{2}{3n} < \frac{2}{3N} < \varepsilon,$$

since $2 + ne^{-n} < 2 + n \le 2n$ for all $n \ge 2$; and $\sin n + 1 \ge -1 + 1 = 0$.

Question 11. Suppose that a sequence (a_n) converges to 0.001. Prove that finitely many values of a_n are negative.

Proof. Given that for any positive real number ε, there exists a natural number N, such that $n > N \Rightarrow \left|a_n - 0.001\right| < \varepsilon$. Choose $\varepsilon = 0.001$, then $n > N \Rightarrow 0 < a_n < 0.002$. Thus, there are finitely many values of a_n are negative, note that $a_n < 0$ for some $n \le N$ might not even exist. Note that maybe some a_n ($n \le N$) may be in the range (0, 0.002) too.

Question 12.

Let (a_n) be a sequence with $a_n \in \mathbb{Z}$ for all natural numbers n.

(a) Is it possible for $\lim(a_n) = 3.5$?

(b) If $a_n \ne a_m$ for all $n \ne m$, prove that (a_n) does not converge.

(c) If (a_n) converges, what can be said about this sequence?

Solution. (a) Suppose $\lim\left(a_n\right) = 3.5$. Then

$$\forall \varepsilon > 0 \ \exists N \in \mathbb{N} \ni n > N \Rightarrow \left|a_n - 3.5\right| < \varepsilon \ \forall n \in \mathbb{N}.$$

Choose $\varepsilon = \frac{1}{4}$ we then have $3.25 = 3.5 - 0.25 < a_n < 3.5 + 0.25 = 3.75$. This is a contradiction because no integer is in between 3.25 and 3.75.

(b) Suppose $a_n \neq a_m$ for all $n \neq m$ but (a_n) converges to a. Then for any positive real number ε, there exists a natural number M, such that

$$n > M \Rightarrow \left|a_n - a\right| < \tfrac{\varepsilon}{2} \text{ and } m > M \Rightarrow \left|a_m - a\right| < \tfrac{\varepsilon}{2}.$$

Then $$0 \underset{\because a_n \neq a_m}{<} \left|a_n - a_m\right| = \left|a_n - a - (a_m - a)\right| \leq \left|a_n - a\right| + \left|a_m - a\right| < \varepsilon.$$

But infinite descent on such numbers is impossible $\because \left|a_n - a_m\right| \in \mathbb{N}$, a contradiction.

(c) $(a_n) = (c)$, a constant sequence, $c \in \mathbb{Z}$.

Question 13. Let $\lfloor x \rfloor$ be the greatest integer $\leq x$ (a floor function). For example, $\lfloor \pi \rfloor = 3$ and $\lfloor a \rfloor = a$ if $a \in \mathbb{Z}$. For each sequence, find $\lim(a_n)$ and verify it with the definition of convergence.

(a) $a_n = \left\lfloor \frac{5}{n} \right\rfloor$, $\lim(a_n) = 0$

(b) $a_n = \left\lfloor \frac{12+4n}{3n} \right\rfloor$, $\lim(a_n) = 1$

Proof. (a) $a_n = \left\lfloor \frac{5}{n} \right\rfloor = 0$ for all $n > 5$. Therefore,

$$n > N > 5 \Rightarrow \left\| \tfrac{5}{n} \right\rfloor - 0 \Big| = 0 < \varepsilon \text{ for any } \varepsilon > 0.$$

(b) $a_n = \left\lfloor \frac{4}{n} + \frac{4}{3} \right\rfloor = 1$ for all $n > 6$. Therefore,

$$n > N > 6 \Rightarrow \left| \left\lfloor \tfrac{4}{n} + \tfrac{4}{3} \right\rfloor - 1 \right| = \left|1 - 1\right| = 0 < \varepsilon \text{ for any } \varepsilon > 0.$$

Question 14. Let $\sigma : \mathbb{N} \to \mathbb{N}$ be a bijection (σ is called a permutation of $\mathbb{N}$). Show that if (a_n) converges to $a \in \mathbb{R}$, then $\left(a_{f(n)}\right)$ also converges to a. (This shows that any reordering of a convergent sequence still converges to the same point.)

Proof. Theorem 9.3: (a_n) converges if and only if (a_n) is Cauchy.

Then for all $\varepsilon > 0$, there exists a $N \in \mathbb{N}$ such that

$$\left|a_n - a_{f(n)}\right| < \tfrac{\varepsilon}{2} \text{ and } \left|a_n - a\right| < \tfrac{\varepsilon}{2} \ \forall n, f(n) > N.$$

It follows that $\left|a_{f(n)} - a\right| = \left|a_{f(n)} - a_n - (a - a_n)\right| \leq \left|a_{f(n)} - a_n\right| + \left|a_n - a\right| < \varepsilon$.